

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

Experiences

Software Engineer

Privex

October 2023 – January 2024

London, United Kingdom

- Developed an AI-driven support assistant using Retrieval-Augmented Generation (RAG) on a self-hosted LLM, reducing staff workload for common inquiries related to technical issues, products, and services.
- Implemented REST APIs in Flask for interfacing with the LLM, integrated with Discord APIs to enable customer queries and on-demand data updates by staff.
- Developed data ingestion and management workflows to process and structure context data from multiple sources, including website documentation, bulk files, support tickets, and Discord announcements.
- Built and maintained infrastructure for efficient data storage and retrieval, using MySQL for chunked data storage and ChromaDB vector database for optimized embedding retrieval.
- Developed systemd service files and monitoring scripts to ensure system reliability, uptime, and automated error recovery.
- Tech Stack:** Python, Flask, ChromaDB, Langchain, MySQL, Linux

Software Engineer

CogniPet

April 2019 – December 2019

Zürich, Switzerland

- Developed a Pet Breed Detection system using the Xception neural network and transfer learning in Keras, achieving 94% accuracy across 120 dog breeds.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).
- Experimented with video-based techniques for a pet facial recognition system by collecting data from dogs and analyzing features from multiple frames.
- Tech Stack:** Python, Flask, Keras, AWS (EC2, S3)

Founder and Developer

TaffyHub.com

August 2017 – July 2019

Tamil Nadu, India

- Developed and launched a school-specific social networking platform for fellow students, reaching over 500+ registered users with 150+ daily active users.
- Implemented key features including news feed, blog publishing system, Q&A system, friend search, and profiling system.
- Built the backend using PHP and MySQL, handling approximately 100 posts per day.
- Configured Nginx reverse proxies for load balancing, caching media files, and used Memcached to speed up query performance.
- Managed full DevOps lifecycle including server setup, automated backups, and system maintenance.
- Made approximately \$100 a month via Google AdSense and membership.
- Tech Stack:** PHP, JavaScript, MySQL, Memcached, Linux (rsync, cron)

Projects

Jix Interpreter | C Interpreter

- Developed an interpreter for a dynamically typed programming language.
- Implemented fault-tolerant parsing and error handling, arrays, function pointers, loops, nested blocks, recursion, built-in functions, conditional statements, and control flow statements.
- Implemented basic functional programming support, including first-class and higher-order functions.
- Tested using automated tests with CI/CD pipelines.
- Tech Stack:** C, CMake, GDB, GitHub Actions

Prism Coverage | Java Code Coverage Engine

- Developed a code coverage engine using ANTLR4 for a programming language.
- Implemented control flow coverage (statement, condition, path) and data-flow coverage (All-Defs, All-DU-Paths, All-P-Uses).
- Implemented color-coded HTML report generation for the coverage engine.
- Tech Stack:** Java, ANTLR4, Maven, JUnit, HTML/CSS

Technical Skills

Languages: Python, JavaScript, Java, C++, C, SQL, PHP, Perl, Bash, HTML/CSS

Technologies: Linux, PostgreSQL, Docker, Kubernetes, MongoDB, Memcached, Redis, AWS, Neo4j

Frameworks: Node.js, Flask, Django, Langchain, ChromaDB

Tools: Git, CI/CD, CMake, Maven, Linux Containers (LXC/LXD)